

Technical Security Assessment & Network Reconnaissance Report

Target Environment Analysis: 10.10.1.0/24

Prepared For	Internal Security Operations & Architecture Teams
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1. Executive Summary

During a routine security infrastructure verification exercise on July 4, 2026, a comprehensive network scanning and passive discovery phase was conducted against selected hosts within the internal subnet range **10.10.10/24**. The primary objective was to map active endpoints, identify running services, ascertain operating system variations, and flag configuration risks that could serve as potential beachheads for an external or insider threat actor.

Key Insights & Findings:

- **Critical Domain Asset Uncovered:** Host 10.10.1.22 was identified as a core Active Directory infrastructure component running **Windows Server 2022 Standard** (Computer Name: Server2022, Domain: CEH.com).
- **Over-Exposed Attack Surface:** The Active Directory Domain Controller exposes multiple sensitive services including SMB, LDAP/s, Kerberos, Microsoft SQL Server (Port 1433), and WS-Man (Port 5985).
- **Network Boundary Detection:** Host 10.10.1.2 was discovered acting as a network gateway/firewall deploying a **pfSense Login Dashboard** on standard HTTP/DNS ports, mapping out routing boundaries.

2. Assessment Methodology

The assessment adhered strictly to standardized technical reconnaissance frameworks, transitioning through the following structured phases:

1. **Host Discovery:** Confirming active targets utilizing local ARP pinging and UDP-based host discovery vectors to bypass basic ICMP filters.
2. **TCP Port Scanning:** Executing complete full three-way handshake TCP Connect scans and stealthy SYN scans across targeted scopes to identify responsive listeners.
3. **Service & OS Fingerprinting:** Interrogating exposed interfaces to elicit system banners, SMB dialect handshakes, and NetBIOS identification packets.
4. **Distributed Sweeping:** Validating open ports globally via modular frameworks (Metasploit Auxiliary Modules) to assure structural replication accuracy.

3. Detailed Target Analysis

Target Host 1: 10.10.1.22 (Primary Domain Controller)

Attribute	Discovered Metric Value
Operating System	Windows Server 2022 Standard 20348 (Build 6.3)
NetBIOS / Computer Name	SERVER2022 \ Server2022.CEH.com
Domain / Forest Name	CEH.com
MAC Address (NIC)	00:15:5D:01:80:02 (Microsoft Corporation Hyper-V)

Active Ports & Exposed Services Profile

Dual-method scanning using full TCP Connect (nmap -sT) and SYN Stealth (nmap -sS) yielded an identically dense service distribution across 23 open ports. The following table identifies primary standard components:

Port/Proto	State	Service Class	Security Context & Operational Risk Implication
53/tcp	open	domain	Active Directory DNS Server. Vulnerable to zone transfers if unhardened.
88/tcp	open	kerberos-sec	Authentication server. Subject to user enumeration and Kerberoasting attacks.
135, 593/tcp	open	msrpc / http-rpc	Microsoft RPC Endpoint Mapper. Leaks endpoint locations and system paths.
389, 636/tcp	open	ldap / ldapssl	Directory entry queries. LDAPS confirmed; plain text LDAP should be audited.
445/tcp	open	microsoft-ds	SMB2/3 Service. Message signing is configured as Required (Secure).
1433/tcp	open	ms-sql-s	Microsoft SQL Server database engine exposed on a critical Domain Controller.
3389/tcp	open	ms-wbt-server	Remote Desktop Protocol (RDP) available; exposes administrative login gateway.
5985/tcp	open	wsman	Windows Remote Management (WinRM HTTP). Target for remote shell access.

SMB Security Architecture Configuration

An advanced script interrogation profile executed via Nmap NSE scripts successfully harvested the security controls active on the SMB layer:

```
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
| smb-os-discovery:
|   OS: Windows Server 2022 Standard 20348
|   Computer name: Server2022
|_  Domain name: CEH.com
```

Note: Finding message signing "enabled and required" indicates a positive baseline posture protecting the host natively from standard SMB Relay/Man-in-the-Middle (MitM) credential manipulation vectors.

Target Host 2: 10.10.1.2 (Network Perimeter Security Appliance)

A wide-scope network sweep utilizing Metasploit network scripts identified an active boundary firewall configuration serving the network ecosystem.

Port/Proto	State	Service Banner	Context
53/tcp	open	Unbound DNS	Secure recursive caching resolver.
88/tcp	open	nginx	Web server hosting administrative components.

```
[msf] >> nmap -Pn -sS -A -oX Test 10.10.1.0/24
```

```
Nmap scan report for 10.10.1.2
Host is up (0.00066s latency).
PORT      STATE SERVICE VERSION
53/tcp    open  domain Unbound
88/tcp    open  http  nginx
|_http-title: pfSense - Login
MAC Address: 02:15:5D:21:A2:2A (Unknown)
```

4. Protocol and Network Traffic Verification

An inspection of internal interface packet streams validated active communications and behavioral anomalies across the local segment.

Packet Level Analysis (Wireshark Capture Highlights)

- **Local Address Resolution Routing:** Multiple MDNS queries originating from host 10.10.1.14 casting out to multicast address 224.0.0.251 flag continuous local asset link discovery.
- **Abrupt Connection Terminations (RST Blocks):** Frame 179 identifies an intentional connection reset sequence:

```
Source: 20.190.148.166 | Destination: 10.10.1.11 | Protocol: TCP
Info: 443 -> 54539 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
```

This trace registers a standard external web resource or authentication endpoint rejecting/closing a socket session initiated by internal node 10.10.1.11.

5. Targeted Remediation Roadmap

To reduce risk vectors surfaced across the identified assets, the following mitigation steps must be prioritized:

Vulnerability Vector	Risk Priority	Targeted Resolution Requirement
Exposed Administrative Interfaces (RDP/ WinRM)	HIGH	Enforce network-level ACLs restricting Port 3389 and Port 5985 access exclusively to dedicated Management Jumpboxes. Do not allow subnet-wide accessibility.
Exposed Production Database Engine	HIGH	Isolate MS-SQL (Port 1433) instance from the core Domain Controller infrastructure. Host database elements on dedicated tier-isolated application instances.
Perimeter Portal Visibility (pfSense HTTP)	MEDIUM	Reconfigure pfSense web configurator to alternate non-standard ports, deactivate HTTP listening, and restrict access natively to localized console interfaces.